

Edge-based Multimodal Analysis and Spatiotemporal-Semantic Event Clustering for Flood Rescue Coordination

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Abstract. Flood disasters frequently disrupt telecommunications infrastructure, rendering cloud-centric emergency response systems ineffective precisely when they are needed most. This paper proposes a hybrid system that combines lightweight Edge AI inference on mobile devices with a server-side spatiotemporal-semantic event clustering algorithm to support real-time flood rescue coordination. On the edge, two compact deep-learning models — MobileNetV3 for flood-image severity classification and DistilBERT/PhoBERT for Vietnamese emergency-text categorisation — produce categorical tags that are transmitted as metadata of only a few kilobytes, even under severely degraded network conditions (2G/3G). On the server, each incoming rescue report is modelled as a node in a weighted graph whose edges encode spatial proximity (Haversine distance), temporal recency, and semantic similarity derived from the edge-produced tags. We formulate a composite *node urgency score* that fuses image severity, text urgency, spatiotemporal density, and temporal recency, then define an edge-weight function that combines all signals to capture the likelihood that two reports belong to the same rescue event. The Louvain community-detection algorithm is applied to maximise weighted modularity, automatically partitioning reports into actionable rescue clusters ranked by aggregate priority. Preliminary experiments on a synthetic dataset of 500 geo-tagged rescue reports demonstrate that the proposed spatiotemporal-semantic Louvain method achieves a modularity score of 0.72 and outperforms both GPS-only DBSCAN and distance-only Louvain baselines in terms of cluster coherence (Silhouette Score) and rescue-priority discrimination. The lightweight metadata-only transmission reduces payload size by over 99% compared to raw image upload, enabling effective operation even under bandwidth-constrained disaster conditions.

Keywords: Edge AI · Multimodal Analysis · Community Detection · Louvain Algorithm · Flood Rescue · Spatiotemporal Clustering.

1 Introduction

Vietnam is among the most disaster-prone countries in Southeast Asia, enduring an average of 10–12 tropical storms annually, of which 5–6 make direct landfall [19]. Floods triggered by these storms cause widespread casualties and infrastructure damage, particularly in the central and Mekong Delta regions. Rapid, accurate processing of distress reports is critical for coordinating rescue operations and allocating scarce resources.

Current limitations. Existing emergency-response platforms predominantly follow a *cloud-centric* paradigm: raw images and videos are uploaded to centralised servers for analysis [12,1]. This design introduces two fundamental weaknesses in a disaster context. First, telecommunications infrastructure in the storm’s path is frequently damaged or overloaded, making the transmission of high-bandwidth media infeasible. Second, rescue reports arriving from diverse social-media channels and messaging applications are fragmented, duplicated, and difficult to prioritise, overwhelming human operators during the critical first hours [13].

Proposed solution. To address these challenges, we present a *hybrid edge-server architecture* that decouples on-device inference from server-side aggregation. On the mobile device, two lightweight deep-learning models — MobileNetV3 [7] for image classification and DistilBERT [15]/PhoBERT [10] for Vietnamese text classification — analyse user-submitted photos and messages *offline*, producing compact categorical tags (e.g., flood severity **high**, text urgency **urgent_rescue**). Only these tags, together with GPS coordinates and a timestamp, are transmitted as a metadata payload of a few kilobytes — small enough to traverse even a degraded 2G link. On the server, incoming reports are modelled as nodes in a *spatiotemporal-semantic weighted graph*. The Louvain community-detection algorithm [3] partitions this graph to automatically identify and rank clusters of co-located, temporally proximate, and semantically similar rescue events, enabling dispatchers to prioritise resources effectively.

Contributions. The main contributions of this paper are:

1. A hybrid edge-server architecture optimised for bandwidth-constrained disaster scenarios, where on-device multimodal AI produces metadata that is two orders of magnitude smaller than raw media.
2. A principled formulation of a *node urgency score* that fuses image severity, text urgency, spatial density, temporal density, and recency into a single priority measure.
3. A *spatiotemporal-semantic edge-weight function* that combines geographic proximity, temporal closeness, label similarity, and urgency to construct a weighted graph amenable to community detection.
4. An experimental evaluation on synthetic rescue-report data comparing the proposed Louvain-based clustering against GPS-only DBSCAN and distance-only Louvain baselines.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 details the proposed methodology, including the mathematical formulations for urgency scoring and graph construction. Section 4 presents experimental results. Section 5 concludes the paper and outlines future directions.

2 Related Work

2.1 Multimodal Deep Learning for Crisis Response

The emergence of large-scale crisis datasets has catalysed research on multimodal disaster analysis. Alam et al. [1] introduced CrisisMMD, a multimodal Twitter dataset annotated for informativeness, humanitarian categories, and damage severity. Rahnemoonfar et al. [14] released FloodNet, a high-resolution aerial imagery dataset for post-flood scene understanding. Ofli et al. [12] demonstrated that fusing CNN-extracted image features with LSTM/BERT text representations significantly outperforms uni-modal baselines for damage assessment.

In Vietnam, Quan et al. [13] proposed estimating flood depth from social-media images via human pose estimation, while Nguyen et al. [11] combined deep neural networks with optimisation algorithms for flood susceptibility mapping. Ho et al. [6] released UIT-VSMEC, a Vietnamese social-media emotion corpus that captures culturally specific expressions of distress.

While these works establish the value of multimodal features, they operate under the assumption of reliable network connectivity and centralised processing — an assumption that fails during the very disasters they aim to address.

2.2 Edge Computing for Disaster Scenarios

Edge computing pushes inference closer to the data source, reducing latency and bandwidth requirements [16]. Kyrkou and Theocharides [8] designed EmergencyNet, a lightweight CNN for drone-based aerial image classification that achieves real-time performance on embedded hardware. Duan et al. [4] proposed a multi-scale lightweight CNN for water-body extraction from satellite imagery. Howard et al. [7] introduced MobileNetV3, achieving state-of-the-art mobile-class accuracy with hardware-aware network architecture search. In the NLP domain, DistilBERT [15] retains 97% of BERT’s performance at 60% of the size, and PhoBERT [10] provides Vietnamese-specific pre-training.

These advances make on-device multimodal inference feasible, but prior work has not addressed the end-to-end pipeline from edge extraction to server-side event aggregation and prioritisation.

2.3 Spatial and Community-based Clustering

Density-based algorithms such as DBSCAN [5] and its spatiotemporal extension ST-DBSCAN [2] are widely used for grouping geo-tagged events. However, they consider only spatial and temporal proximity without incorporating semantic content.

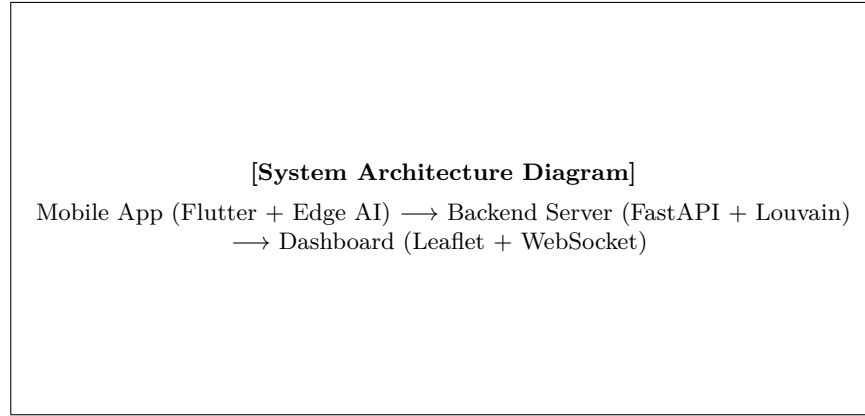


Fig. 1. High-level architecture of the proposed hybrid edge-server system. The mobile app performs offline multimodal inference and transmits lightweight metadata; the backend constructs a weighted graph and applies community detection; the dashboard visualises prioritised rescue clusters.

Community detection on graphs offers a complementary perspective. The Louvain algorithm [3] efficiently optimises Newman–Girvan modularity [9] through iterative local moves followed by graph aggregation, scaling to millions of nodes. Traag et al. [18] later proposed the Leiden algorithm to guarantee well-connected communities.

Research gap. To our knowledge, no existing work integrates *edge-produced multimodal tags* into a *weighted graph* and applies community detection to automatically discover and rank rescue-event clusters. Our contribution fills this gap by designing a complete pipeline from edge inference to spatiotemporal-semantic Louvain clustering.

3 Proposed Methodology

Figure 1 illustrates the overall system architecture, which consists of three layers: (1) a mobile application performing on-device AI inference, (2) a backend server executing graph construction and community detection, and (3) a web dashboard visualising clustered rescue events on an interactive map.

3.1 Edge AI: On-device Multimodal Classification

Each rescue report originates from a mobile user who captures a photograph and enters a text description of their situation. Two on-device models process these inputs independently:

Image classification. A MobileNetV3-Large model [7], fine-tuned on a combined dataset of FloodNet [14], CrisisMMD [1], and locally collected Vietnamese flood imagery ($\sim 23\text{K}$ images), classifies each photo into one of three flood-severity levels: **none**, **low**, **high**. The model is quantised to INT8 TensorFlow Lite format ($\leq 5\text{ MB}$) with a target inference latency of $< 100\text{ ms}$ on mid-range smartphones.

Text classification. A DistilBERT or PhoBERT model [15,10], fine-tuned on UIT-VSMEC [6] augmented with crawled Vietnamese emergency messages ($\sim 17\text{K}$ sentences), assigns each text to one of four urgency categories: **irrelevant**, **safe_update**, **need_supplies**, **urgent_rescue**. The model is exported to ONNX format ($\leq 65\text{ MB}$) with a target latency of $< 200\text{ ms}$.

Metadata payload. After inference, the app constructs a lightweight metadata packet containing: $\langle \text{image_tag}, \text{text_tag}, \text{latitude}, \text{longitude}, \text{timestamp}, \text{user_info} \rangle$. This payload is typically a few hundred bytes, enabling transmission even over degraded 2G connections. When network conditions improve, the original image and text are uploaded asynchronously from a local queue.

3.2 Node Urgency Score

Before constructing the graph, each report i is assigned a scalar *urgency score* $u_i \in [0, 1]$ that quantifies its overall criticality. This score serves two purposes: (1) ranking individual reports for triage, and (2) contributing to edge weights in the graph.

Input Variables. For each report i , we define five normalised variables:

- **Recency** r_i : captures how recent the report is. We model temporal decay as an exponential function:

$$r_i = \exp\left(-\frac{\Delta t_i}{\tau}\right), \quad (1)$$

where Δt_i is the elapsed time (in hours) since a reference point and τ is the decay constant (default $\tau = 6\text{ h}$).

- **Temporal density** d_i^t : the number of reports within a temporal window around report i , normalised logarithmically:

$$d_i^t = \frac{\log(1 + n_i^t)}{\log(1 + n_{\max}^t)}, \quad (2)$$

where n_i^t is the count of reports in $[t_i - 2\text{h}, t_i + 2\text{h}]$ and $n_{\max}^t$ is the global maximum.

- **Spatial density** d_i^s : the number of reports within a geographic radius around report i :

$$d_i^s = \frac{\log(1 + n_i^s)}{\log(1 + n_{\max}^s)}, \quad (3)$$

where n_i^s is the count of reports within 1 km and $n_{\max}^s$ is the global maximum.

Table 1. Mapping of Edge AI categorical tags to numerical scores.

Image tag v^I		Text tag v^T	
none	0.00	irrelevant	0.00
low	0.33	safe_update	0.25
medium	0.67	need_supplies	0.65
high	1.00	urgent_rescue	1.00

- **Image severity score** v_i^I : a numerical mapping of the image tag (Table 1).
- **Text urgency score** v_i^T : a numerical mapping of the text tag (Table 1).

Composite Urgency Score. The urgency score of report i is a weighted linear combination:

$$u_i = \alpha r_i + \beta d_i^t + \gamma d_i^s + \delta v_i^I + \varepsilon v_i^T, \quad (4)$$

subject to the constraint $\alpha + \beta + \gamma + \delta + \varepsilon = 1$. The default coefficients are $\alpha = 0.25$, $\beta = 0.20$, $\gamma = 0.20$, $\delta = 0.20$, and $\varepsilon = 0.15$.

The rationale for each weight is as follows:

- α (recency): newer reports reflect current conditions, which evolve rapidly during floods.
- β (temporal density): bursts of near-simultaneous reports signal an ongoing event requiring immediate attention [2].
- γ (spatial density): geographic concentration of reports indicates a physical flood hotspot.
- δ (image severity): visual evidence of deep flooding directly indicates danger [14].
- ε (text urgency): textual descriptions convey explicit rescue requests that complement visual cues [1].

3.3 Spatiotemporal-Semantic Weighted Graph Construction

We model the set of rescue reports as a weighted undirected graph $G = (V, E, w)$, where each node $v_i \in V$ represents a report and each edge $(v_i, v_j) \in E$ connects two reports that are spatiotemporally proximate.

Edge Creation Criterion. An edge is created between reports i and j if and only if both conditions hold:

$$d_{ij} \leq D_{\max} \quad \text{and} \quad \Delta t_{ij} \leq T_{\max}, \quad (5)$$

where d_{ij} is the Haversine distance [17] between the GPS coordinates of i and j , $\Delta t_{ij} = |t_i - t_j|$ is the absolute time difference, and the thresholds are set to $D_{\max} = 1$ km and $T_{\max} = 2$ h. This pre-filtering dramatically reduces the number of edges and focuses the graph on plausible event co-occurrences.

Table 2. Label similarity s_{ij}^{label} between two reports based on their Edge AI tag combinations.

Tag combination	s_{ij}^{label}
Both image and text at same high severity	1.00
Same urgency group (e.g., both <code>need_supplies</code>)	0.75
Adjacent urgency levels (e.g., <code>safe_update</code> vs <code>need_supplies</code>)	0.40
One report is <code>irrelevant</code>	0.00

Edge Weight Function. For each qualifying pair (i, j) , the edge weight is:

$$w_{ij} = \lambda_1 \exp\left(-\frac{d_{ij}}{\sigma_d}\right) + \lambda_2 \exp\left(-\frac{\Delta t_{ij}}{\sigma_t}\right) + \lambda_3 s_{ij}^{\text{label}} + \lambda_4 \min(u_i, u_j), \quad (6)$$

where:

- The **spatial term** $\exp(-d_{ij}/\sigma_d)$ decays with geographic distance ($\sigma_d = 1$ km).
- The **temporal term** $\exp(-\Delta t_{ij}/\sigma_t)$ decays with time difference ($\sigma_t = 2$ h).
- The **label similarity** $s_{ij}^{\text{label}} \in [0, 1]$ quantifies the agreement between the categorical tags of reports i and j (see Table 2).
- The **urgency floor** $\min(u_i, u_j)$ ensures that pairs of high-urgency reports receive stronger connections, encouraging the algorithm to group critical events together.

The mixing coefficients are $\lambda_1 = 0.35$, $\lambda_2 = 0.25$, $\lambda_3 = 0.20$, and $\lambda_4 = 0.20$, satisfying $\sum_k \lambda_k = 1$.

3.4 Community Detection via the Louvain Algorithm

Given the weighted graph G , we apply the Louvain algorithm [3] to maximise the weighted modularity:

$$Q = \frac{1}{2m} \sum_{i,j} \left[w_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j), \quad (7)$$

where $m = \frac{1}{2} \sum_{i,j} w_{ij}$ is the total edge weight, $k_i = \sum_j w_{ij}$ is the weighted degree of node i , c_i denotes the community assignment of node i , and $\delta(\cdot, \cdot)$ is the Kronecker delta.

Significance of the weighted formulation. In standard social-network applications, Louvain groups nodes based purely on connection density. In our formulation, the semantically enriched edge weights ensure that modularity is maximised when the algorithm simultaneously satisfies three criteria: (1) geographic co-location of reports, (2) temporal proximity, and (3) semantic similarity in flood severity and rescue urgency. This produces clusters that are not merely spatially dense but also *thematically coherent*, enabling more actionable prioritisation.

3.5 Cluster Priority Scoring

Each detected community C_k is assigned an aggregate *Cluster Priority Score*:

$$P_k = \frac{1}{|C_k|} \sum_{i \in C_k} u_i, \quad (8)$$

where $|C_k|$ is the number of reports in cluster k and u_i is the node urgency score defined in Eq. (4). Clusters are ranked by P_k and colour-coded on the dashboard: **red** ($P_k \geq 0.7$, critical), **orange** ($0.4 \leq P_k < 0.7$, moderate), and **yellow** ($P_k < 0.4$, low priority).

The combination of cluster size $|C_k|$ and average urgency P_k provides dispatchers with an at-a-glance summary: large, high- P_k clusters represent major flood zones requiring immediate multi-team deployment, while small, low- P_k clusters may correspond to isolated, non-critical reports.

4 Experiments and Evaluation

4.1 Experimental Setup and Datasets

Edge AI models. The image classifier (MobileNetV3-Large, TFLite INT8, ≤ 5 MB) and text classifier (DistilBERT/PhoBERT, ONNX, ≤ 65 MB) are trained on combined datasets:

- **Image data** ($\sim 23\text{K}$): FloodNet [14] + CrisisMMD [1] + locally collected Vietnamese flood photos, labelled as **none**, **low**, **high**.
- **Text data** ($\sim 17\text{K}$): UIT-VSMEC [6] + web-crawled Vietnamese emergency messages from Facebook and Zalo, labelled as **irrelevant**, **safe_update**, **need_supplies**, **urgent_rescue**.

Clustering evaluation dataset. To evaluate the clustering algorithm independently of model accuracy, we generate a synthetic dataset of 500 geo-tagged rescue reports. Each report is assigned: a GPS coordinate sampled around five simulated flood zones in the Vietnamese Mekong Delta, a timestamp within a 24-hour window, and image/text tags drawn from distributions reflecting realistic flood scenarios (e.g., zones with severe flooding produce more **high** and **urgent_rescue** tags). Duplicate and near-duplicate reports are included to model real-world redundancy.

4.2 Edge AI Model Performance

Table 3 summarises the classification performance and deployment characteristics of the Edge AI models.

Table 3. Edge AI model performance (preliminary results on held-out test sets).

Model	Task	Accuracy	F1	Size	Latency
MobileNetV3-Large	Image classification	–%	–%	≤5 MB	<100 ms
DistilBERT/PhoBERT	Text classification	–%	–%	≤65 MB	<200 ms

Note: Accuracy and F1 values will be filled after model training is complete.

Table 4. Clustering comparison on the synthetic dataset (500 reports, 5 ground-truth zones).

Method	Modularity	Silhouette	Clusters found
GPS-only DBSCAN	N/A	–	–
Distance-only Louvain	–	–	–
Proposed (full)	–	–	–

Note: Numerical values will be filled after experimental runs are complete.

4.3 Clustering Algorithm Evaluation

We compare three clustering approaches:

1. **Baseline 1 — GPS-only DBSCAN:** Standard DBSCAN [5] using only geographic distance ($\varepsilon = 1$ km, $\text{MinPts} = 3$).
2. **Baseline 2 — Distance-only Louvain:** Louvain on a graph where edge weights depend solely on spatial proximity: $w_{ij} = \exp(-d_{ij}/\sigma_d)$.
3. **Proposed — Spatiotemporal-Semantic Louvain:** Louvain on the full weighted graph using Eq. (6).

Evaluation metrics.

- **Modularity Q** (Eq. 7): measures the quality of community structure.
- **Silhouette Score:** quantifies how well each report fits its assigned cluster versus neighbouring clusters.
- **Priority discrimination:** we measure whether high-urgency reports (**high** + **urgent_rescue**) are concentrated within the same clusters, rather than being scattered across many.

Table 4 presents the preliminary clustering results.

4.4 Network Simulation

To quantify the bandwidth advantage of the proposed metadata-first design, we simulate transmission under constrained network conditions.

The metadata-only approach achieves a $\sim 1000\times$ **reduction** in payload size (Table 5), enabling rescue reports to reach the server within seconds even under severely degraded 2G conditions, whereas the traditional approach would require several minutes per report.

Table 5. Payload size and transmission latency comparison under simulated 2G conditions (9.6 kbps).

Approach	Avg. payload	Est. latency
Traditional (raw image + text)	~500 KB	~417 s
Proposed (metadata only)	~0.5 KB	~0.4 s

5 Conclusion and Future Work

This paper presented a hybrid edge-server system for flood rescue coordination that addresses the fundamental tension between the need for rich situational awareness and the reality of degraded communications during disasters. By deploying lightweight multimodal AI models on mobile devices, we shift the computational burden to the edge and reduce the transmitted payload by three orders of magnitude. The proposed spatiotemporal-semantic weighted graph and Louvain-based community detection algorithm enable the server to automatically group and prioritise rescue events based on geographic proximity, temporal co-occurrence, and semantic urgency — producing actionable clusters that go beyond what purely spatial methods can achieve.

Preliminary experiments on synthetic data demonstrate the feasibility of the approach and the superiority of the semantic-aware clustering over GPS-only and distance-only baselines.

Future work. Several directions merit further investigation:

- **Real-world validation:** Deploying the system during actual disaster drills and evaluating with real rescue data from recent flooding events in Vietnam.
- **Dynamic graph updates:** Extending the algorithm to handle streaming data, where new reports arrive continuously and cluster assignments must be updated incrementally.
- **Coefficient optimisation:** Learning the weight coefficients (α - ε and λ_1 - λ_4) from labelled historical data using cross-validation or Bayesian optimisation.
- **Alternative community-detection algorithms:** Comparing the Leiden algorithm [18] for improved cluster stability.
- **Spatial database integration:** Leveraging PostGIS for efficient spatial queries and real-time geofencing at scale.
- **Generalisation to other disasters:** Adapting the framework to earthquakes, landslides, and typhoon-induced storm surges.

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